

PAX Purview Audit Log Processor v1.11.3 Changelog

Release Date: TBD Previous Version: v1.11.2

Summary

Trims the `-IncludeM365Usage` activity bundle from roughly one hundred operations down to a curated twenty-two-operation set focused on Exchange mail access, SharePoint/OneDrive file access, Teams chat/messaging, Teams meeting lifecycle, and Copilot/Connected-AI signals — the shape consumed by the Analytics-Hub M365 Usage Analytics dashboard. Auto-suppresses the DSPM-for-AI informational Y/N prompt on `-IncludeM365Usage` runs that do not include `AIAppInteraction`, since `ConnectedAIAppInteraction` is now part of the curated bundle and the prompt is redundant in that context. Refreshes the embedded M365 Bundle Explosion processor to a newer schema (adds four agent-context columns to the rollup) and rewires the M365 `-AppendFile` flow so the union-merge and sidecar regeneration are driven from PowerShell rather than from inside the embedded processor; the customer-visible append behavior under `-IncludeM365Usage + -Rollup / -RollupPlusRaw` is preserved end-to-end (the customer’s target Rollup is updated in place, and the two sidecars overwrite the same destination URLs on every run). Loosens the v1.11.2 managed-identity host guard so that `-Auth AppRegistration` runs with bound credentials are no longer blocked when the host happens to expose a managed-identity token endpoint. Fixes two Fabric/OneLake destination bugs that surfaced under `-OutputPath` pointing at a lakehouse: the Fabric URL validator now matches the documented “point at the lakehouse root” guidance, and the OneLake upload helper no longer corrupts hierarchical artifact paths during checkpoint mirroring. A pass of cosmetic comment cleanup in the script source, plus a help-block correction that removes a stale `-AppendFile` entry from the `-Rollup` blocked-combinations list (documentation only, no runtime change). No new top-level switches, no removed switches, no dependency changes.

Changes

-IncludeM365Usage activity bundle trimmed to a curated 22-operation set

The activity-type bundle attached by `-IncludeM365Usage` is trimmed from roughly one hundred operations down to a curated twenty-two-operation set targeted at the Analytics-Hub M365 Usage Analytics dashboard. The trimmed bundle is, in declaration order:

Workload group	Operations
Exchange / Email	<code>MailItemsAccessed</code> , <code>MailboxLogin</code> , <code>Send</code>
SharePoint / OneDrive (file access)	<code>FileAccessed</code> , <code>FileViewed</code> , <code>FilePreviewed</code> , <code>FileModified</code> , <code>FileDownloaded</code> , <code>FileUploaded</code>
Teams (chat / messaging)	<code>MessageSent</code> , <code>MessageRead</code> , <code>MessagesListed</code> , <code>ChatRetrieved</code> , <code>ChatCreated</code> , <code>TeamsSessionStarted</code>
Teams (meeting lifecycle)	<code>MeetingParticipantJoined</code> , <code>MeetingStarted</code> , <code>MeetingEnded</code> , <code>MeetingParticipantDetail</code> , <code>MeetingDetail</code>
Copilot / Connected AI	<code>CopilotInteraction</code> , <code>ConnectedAIAppInteraction</code>

Three operations are net-new to the bundle relative to v1.11.2 (`MailboxLogin`, `FileViewed`, `FilePreviewed`), one is net-new under the Copilot grouping (`ConnectedAIAppInteraction`), and the broader sets covering Office-app document open/save/print, file sharing, Teams team/channel/app management, Forms/Stream/Planner/PowerApps lifecycle events,

full meeting-recording lifecycle, and unified-group membership changes are removed from the bundle. None of the removed operations are blocked at the parameter layer — any of them can still be requested explicitly via `-ActivityTypes` (with or without `-IncludeM365Usage`), and explicit `-ActivityTypes` continues to union with the bundle on the same run.

The `-IncludeM365Usage` `RECORD TYPES INCLUDED` set is unchanged. The trimmed activity list narrows server-side filtering (Operations and RecordType are AND-ed by `Search-UnifiedAuditLog`), so the bundle now returns exactly the curated operation set against the same record-type universe as v1.11.2. The `-AppendFile` union-merge, the sidecar regeneration, the rollup column schema, the Fabric/OneLake destination handling, and the leaf-shape validator are all untouched by this trim. The in-script `.PARAMETER IncludeM365Usage` help block is updated to list the new operations.

Managed-identity host guard now considers bound credentials

The v1.11.2 guard that fail-fasts a run when `$env:IDENTITY_ENDPOINT` is set but `-Auth` is not `ManagedIdentity` now also accepts `-Auth AppRegistration` when `-ClientId` is bound together with one of `-ClientSecret`, `-ClientCertificateThumbprint`, or `-ClientCertificatePath`. This unblocks the cross-tenant pattern in which a script running on an Azure-hosted runtime (Functions, App Service, Container Apps, Container Instances, Fabric notebook / container) authenticates into a different tenant with explicit App Registration credentials. The guard still fires — same exit code, same overall intent — when `-Auth` is neither `ManagedIdentity` nor a credential-complete `AppRegistration`. The remediation text in the error message is updated to mention both accepted shapes.

Embedded M365 Bundle processor refreshed

The embedded M365 Bundle Explosion processor is refreshed to a newer schema. The aggregated Rollup now carries fourteen columns:

```
UserId, CreationDate, Operation, Workload, SourceFileExtension, AppHost, EventCount, ItemsAccessedCount,
CreationTime, MaxCreationTime, AgentId, AgentName, ContextType, IsAgentInteraction
```

The four trailing columns (`AgentId`, `AgentName`, `ContextType`, `IsAgentInteraction`) are new in this release. They surface the agent-context attribution carried in the underlying `CopilotInteraction` stream, and they participate in the rollup group key so that distinct agent attributions are no longer collapsed into a single aggregated row.

The processor's output filenames now carry a fresh current-run timestamp suffix on each of the three outputs (Rollup, UserStats, SessionCohort) — for example, `<purview_stem>_Rollup_<YYYYMMDD_HHMMSS>.csv`. The end-of-run local listing and the remote upload sweep both pick these up unchanged (the filenames still contain the run timestamp inherited from the input stem).

M365 -AppendFile union-merge is now PowerShell-driven

The M365 `-AppendFile` flow used to push the customer's target Rollup into the embedded processor and let the processor perform the union-merge in-process. That round-trip is replaced by a dedicated PowerShell merge step (`Merge-M365RollupCsv`) that runs after the embedded processor emits its timestamped Rollup. The merge is column-tolerant — legacy targets that pre-date the 14-column schema (for example, files carried over from a 9- or 10-column rollup produced by an earlier processor build) are accepted as-is and padded with empty `AgentId` / `AgentName` / `ContextType` cells and `IsAgentInteraction=FALSE` during the union, so the merged output retains a uniform 14-column shape end-to-end. The 9-tuple group key (`UserId`, `CreationDate`, `Operation`, `Workload`, `SourceFileExtension`, `AppHost`, `AgentId`, `AgentName`, `ContextType`) and the aggregation semantics (sum / sum / min / max / OR / first-seen casing) match what the embedded processor itself uses, so the union is identical to a fresh full-content rollup over the same input universe.

After the merge writes the union back to the customer's target file, the embedded processor is re-invoked against the merged target to regenerate the two sidecars from the merged content. The sidecars are then anchor-renamed onto the

target leaf stem (the timestamp suffix is dropped on the sidecar leaves) so the destination URLs are overwritten in place. The customer's end state on the destination is the same shape as before: exactly one Rollup, one UserStats, and one SessionCohort at stable URLs.

-AppendFile leaf validation widened for timestamped Rollup leaves

Under `-IncludeM365Usage` with `-Rollup` / `-RollupPlusRaw`, the `-AppendFile` leaf validator previously required the leaf to end in `_Rollup.csv` (anchored, no timestamp). The validator now also accepts `_Rollup_<YYYYMMDD_HHMMSS>.csv`, which is the natural shape of the embedded processor's current-run output. Customers can therefore point `-AppendFile` at a previous run's raw Rollup without first renaming it to drop the timestamp. The same widening applies to the leaf-specific rejection messages for sidecar (`_UserStats`, `_SessionCohort`) and wrong-shape (`_Interactions`, `_Exploded`) inputs. Sidecar leaves remain rejected as merge anchors regardless of whether they carry a timestamp suffix.

DSPM for AI informational prompt auto-suppressed under -IncludeM365Usage

The DSPM-for-AI Billing Information block previously printed an informational banner and a Y/N prompt whenever `$finalActivityTypes` contained any of `AIInteraction`, `ConnectedAIAppInteraction`, or `AIAppInteraction`, unless `-Force` was passed. Now that `ConnectedAIAppInteraction` is part of the curated `-IncludeM365Usage` bundle, every `-IncludeM365Usage` run was tripping that prompt even though the user did not type the DSPM activity type directly. The prompt is now auto-suppressed when `-IncludeM365Usage` is set AND `AIAppInteraction` is not in `$finalActivityTypes`. A short informational line is emitted in place of the prompt:

```
DSPM for AI: -IncludeM365Usage active and AIAppInteraction not requested; auto-continuing without prompt.
```

The PAYG path is unchanged. If a customer explicitly combines `-IncludeM365Usage` with `-ActivityTypes` `AIAppInteraction`, the full DSPM banner and the three-way PAYG Y/N/E prompt still fire (still bypassable only via `-Force`), because `AIAppInteraction` carries real billing implications. The `-Force` semantics for the informational branch are unchanged: `-Force` continues to skip the informational prompt regardless of `-IncludeM365Usage`. The `.PARAMETER Force` help block is updated to mention the new auto-suppression path alongside the existing `-Force` auto-acceptance behavior.

Help-block corrections (`-Rollup`, `-RollupPlusRaw`)

The `.PARAMETER Rollup` help block and the matching `# Rollup post-processor:` comment in the param block listed `-AppendFile` among the blocked combinations under `-Rollup` / `-RollupPlusRaw`. That entry did not match the runtime gating, which has never blocked `-AppendFile`. The stale `-AppendFile` entry is removed. The corrected blocked-combinations list — `-UseEOM`, `-ExportWorkbook`, `-OnlyUserInfo`, `-OnlyAgent365Info`, `-RAWInputCSV`, and `-ExcludeCopilotInteraction` without `-IncludeM365Usage` — matches the validator at runtime. No runtime behavior changes; this entry is help/documentation only.

Fabric / OneLake destination fixes

Two destination-side bugs that surfaced under `-OutputPath` pointing at a Fabric Lakehouse URL are corrected. The fixes together unblock `-IncludeM365Usage` (and any other Fabric-tier run that performs a checkpoint mirror) against a lakehouse destination.

Fabric URL acceptance widened to match documented usage. The Fabric URL validator (`Resolve-FabricTarget`) previously rejected URLs that did not include an explicit `/Files` third segment, which contradicted the in-line help and banner text that recommend pointing `-OutputPath` at the lakehouse root and letting the script route operational artifacts to `Files/` and Delta tables to `Tables/` automatically. The validator now accepts three URL shapes and reports enough metadata for the upload helpers to route correctly:

Shape	Example	Routing
Lakehouse root	<code>https://onelake.dfs.fabric.microsoft.com/<ws>/<item>.Lakehouse</code>	Operational artifacts auto-route under <code>Files/</code> ; Delta tables under <code>Tables/</code>
Explicit Files	<code>https://onelake.dfs.fabric.microsoft.com/<ws>/<item>.Lakehouse/Files[/<rel>]</code>	<code>Files/<rel>/...</code>
Explicit Tables	<code>https://onelake.dfs.fabric.microsoft.com/<ws>/<item>.Lakehouse/Tables[/<schema>]</code>	<code>Tables/<schema>/...</code> (table-typed destinations)

A URL whose third segment is neither `Files` nor `Tables` is still rejected, with a more specific error message. `.Warehouse` items continue to be accepted alongside `.Lakehouse`.

OneLake upload path is now per-segment URI-escaped, and parent directories are pre-created. `Send-FileToOneLake` and `Get-RemoteFile-OneLake` previously escaped a path-shaped `RemoteFileName` / `RelativeName` (for example, `.pax_resume/<RunTimestamp>/<file>` written by `Sync-FabricResumeMirror`) through `[System.Uri]::EscapeDataString` as a single data segment. That turned the embedded `/` separators into `%2F`, malforming the DFS URL so the three-phase Create / Append / Flush calls could land on inconsistent resolved paths — surfacing on the very first checkpoint mirror as:

```
400 (The uploaded data is not contiguous or the position query parameter value
    is not equal to the length of the file after appending the uploaded data.)
```

Both helpers now split the relative name on `/`, escape each segment individually with `[System.Uri]::EscapeDataString`, and re-join with `/`, so the resulting URL preserves the hierarchical path. `Send-FileToOneLake` additionally performs an idempotent `PUT ?resource=directory` against each missing parent directory before the Create File call (a 409 from the directory create is treated as success), so artifacts mirrored to a fresh subfolder (`.pax_resume/<RunTimestamp>/`, log subdirectories, and so on) no longer require the customer to pre-create the layout in OneLake.

The fixes are transparent at the customer surface: no new switches, no behavioral change for the (previously working) `-OutputPath ".../Files/<existing-folder>"` shape, and no schema change to checkpoint or mirror artifacts. A focused Pester gate (`tests\Purview.Fabric.UrlEscape.Tests.ps1`) asserts the URL acceptance contract and the per-segment-escape invariant so a future regression of either fix fails statically.

Comment cleanup (cosmetic only)

In-source comments that referenced historical development markers were rewritten as plain descriptive commentary. Technical content of the comments (concurrency invariants, schema notes, ordering constraints, dedup keys, scratch-vs-remote URL handling, and so on) is preserved verbatim — only the marker framing was removed. No code logic, control flow, function signatures, output messages, log lines, schema columns, or operator-visible behavior is changed by this pass.

Removed

(None.)

Compatibility

- No new top-level switches, no removed switches, no changed switch semantics.
- No new dependencies, no removed dependencies, no changed dependency versions.
- No checkpoint schema changes — checkpoints written by v1.11.2 resume cleanly under v1.11.3.
- **M365 rollup schema is widened from ten to fourteen columns** (added `AgentId`, `AgentName`, `ContextType`, `IsAgentInteraction`). The four added columns are appended at the end, preserving the order of all prior columns; readers that bind by column index or by column name and ignore extras keep working unmodified.
- Cross-version `-AppendFile` merge is supported in both directions. A legacy target file (9- or 10-column) merged with a current-run Rollup is padded to 14 columns during the union; a current-run target file (14-column) merged with itself remains 14 columns.
- M365 rollup output filenames now carry a current-run timestamp suffix (`<stem>_Rollup_<YYYYMMDD_HHMMSS>.csv` and likewise for the two sidecars). The leaf-shape validator accepts both the anchored and the timestamped Rollup shapes as `-AppendFile` targets.
- `-IncludeM365Usage` activity bundle is trimmed from roughly one hundred operations to a curated twenty-two-operation set (Exchange mail access, SharePoint/OneDrive file access, Teams chat/messaging, Teams meeting lifecycle, Copilot/Connected-AI). Operations dropped from the bundle (Office-app open/save/print, file sharing, Teams team/channel/app/approval management, Forms/Stream/Planner/PowerApps lifecycle, full meeting-recording lifecycle, unified-group membership changes) are still queryable by passing them explicitly via `-ActivityTypes`. Three operations (`MailboxLogin`, `FileViewed`, `FilePreviewed`) and one Copilot-grouped operation (`ConnectedAIAppInteraction`) are net-new to the bundle relative to v1.11.2. The `RECORD TYPES INCLUDED` set is unchanged.
- **DSPM-for-AI informational Y/N prompt is auto-suppressed under** `-IncludeM365Usage` when `AIAppInteraction` is not in `$finalActivityTypes`. A single informational line replaces the prompt. The PAYG (`AIAppInteraction`) path and the `-Force` semantics are unchanged; the prompt still fires for explicit `-ActivityTypes AIAppInteraction` runs unless `-Force` is set.
- **Fabric `-OutputPath` URL acceptance is widened, not narrowed.** Every URL shape that v1.11.2 accepted (`.../Files[<rel>]`) continues to be accepted with identical routing. Additionally, the lakehouse-root form (`.../<item>.Lakehouse`) and the explicit Tables form (`.../<item>.Lakehouse/Tables[<schema>]`) are now accepted, matching the in-line help and banner text. A URL whose third segment is neither `Files` nor `Tables` is still rejected.
- OneLake checkpoint-mirror artifacts under `.pax_resume/<RunTimestamp>/` are now written with hierarchical paths preserved on the wire. A run that was previously aborted at the first checkpoint write by a 400 “uploaded data is not contiguous” response from the DFS endpoint now completes against the same destination URL without operator intervention. There is no migration step for any pre-existing local checkpoint, partial output, or `.pax_incremental/*.jsonl` shard.
- No `fabric_resources/` companion changes.